

// Day 44 of 60

ARCHITECTURE

How WhatsApp Handles Billions

→ Swipe through all 6 slides



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The Secret Sauce: Erlang

Built for telecom-grade scale

Why Erlang?

- Built for telecom switches
- Millions of concurrent connections per box
- Lightweight processes
- Crash-resistant by design

```
// 50 engineers handled  
// 900M users at acquisition.
```



Right tool for right problem



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Real-Time WebSockets

No polling, ever

Open app

- WebSocket established
- connection stays alive
- server pushes messages

No polling.

No wasted bandwidth.

Messages arrive < 100ms.

// Persistent connection

// = instant delivery



Polling wastes battery and bandwidth



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Offline Message Queue

Friend offline? Queue it.

Friend offline:

msg → server queue

↓ (waits)

friend reconnects

↓

queue flushes to phone

↓

queue deletes msg

// Server-side storage

// is temporary by design



Messages live on your device, not theirs



End-to-End Encryption

Servers never see content

On your phone:

```
msg + receiver_pub_key  
→ encrypted blob  
→ sent to server
```

Server sees:

```
encrypted blob (opaque)
```

Receiver decrypts locally.

```
// Powered by Signal Protocol
```



Zero-knowledge architecture



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Media Storage Strategy

Photos don't go through chat

Sending a photo:

1. Upload to Blob Storage (S3-style)
2. Get back a URL
3. Send URL + thumbnail through chat socket
4. Receiver downloads from blob storage

// Keeps chat socket light



Never send big payloads through hot paths



Stateless Servers

Scale up, scale down — instantly

App servers store:

- connection map
(and nothing else)

State lives in:

- Database
- Message queues
- User devices

```
// Add 1000 servers in 5 min  
// for New Year's Eve spike
```



Statelessness = horizontal scaling



done = true

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